The Fibonacci sequence is closely related to the number 1.618034, which is known as **phi** (say "fie"). Mathematicians and artists have known about this very peculiar number for several thousand years, and for a long time people thought it had magical properties.

Leonardo da Vinci called phi the "golden ratio" and used it in paintings

the Golden RATIO

Golden spirals

If you draw a rectangle with sides 1 and phi units long, you'll have what artists call a "golden rectangle" – supposedly the most beautiful rectangle possible. Divide this into a square and a rectangle (like the red lines here), and the small rectangle is yet another golden rectangle. If you keep doing this, a spiral pattern begins to emerge. This "golden spiral" looks similar to the shell of a sea creature called a nautilus, but in fact they aren't quite the same. A nautilus shell gets about phi times wider with each half turn, while a golden spiral gets phi times wider with each quarter turn.

Golden rectangles create a spiral that continues forever

Golden rectangle